

Task-matched baselines reveal the limits of computational chart selection

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Abstract

A chart selector should improve on a simple task-matched default, not merely beat displays that omit the requested information. We test this requirement in a rendered-image benchmark of 127 two-group comparisons from nine public resources. An initial five-quantile task favors a direct five-marker display in 122 comparisons, exposing the weakness of mean-only controls. We then evaluate six quantile-based tasks with ten fixed configurations, including both five- and 19-marker controls, under two computational observers. Resource-disjoint task defaults and case-specific selections are frozen at 256 pixels and evaluated at 384 pixels with new placements. Case-specific selection does not consistently beat the task default: its threshold-observer point estimate improves only for tail asymmetry, and that improvement is uncertain under the alternative observer. The dense quantile control belongs to the 19-quantile winning set in 127 and 126 comparisons under the two observers, respectively. We additionally distinguish information omitted by an encoding from raster recovery error, and demonstrate why marginal charts cannot identify median within-person change. The contribution is an executable evaluation of when chart optimization adds value and when its apparent benefit follows from task or decoder choices. These retrospective computational results do not establish human reading accuracy or generalization to an untouched population.

Introduction

A chart should be evaluated against the question its reader is expected to answer. Comparing two means, comparing distributional shapes and estimating individual change are different tasks. Displays that omit the required information cannot become adequate simply through higher resolution. Conversely, retaining observations does not establish that a reader can recover the relevant information accurately. Distributional comparisons and shift functions have long provided alternatives to mean-only summaries.^{1;2} Scientific visualization requires explicit objectives and evidence that the resulting displays support those objectives.^{3;4}

Automated graphical design, constraint-based systems and learned recommenders address chart construction and recommendation.⁵⁻⁷ Task taxonomies clarify the analytic operations a visualization should support, while graphical-perception experiments evaluate human judgments.^{8;9} These contributions motivate, but do not validate, a separate computational question: given known data and a specified recovery algorithm, which rendering most accurately communicates a declared numerical target?

A score alone does not answer whether the recommended chart is useful. A color-calibrated decoder may recover faint heatmap intensities more accurately than a person. A mean-only baseline can be predictably weak for a distributional task. Searching more settings in one family can increase its opportunity to win. Nearly identical scores may exchange rank with a small change in resolution. These concerns require explicit controls rather than a transformation that forces every chart to have a unique score.

We ask a falsifiable question: after admitting a strong task-matched default, does case-specific selection reduce recovery error at a display condition excluded from selection? The benchmark distinguishes

three issues: whether the encoding retains the requested information, how accurately a computational observer recovers it, and whether optimization improves on a fixed rule. Its contributions are an explicit information-eligibility check, a matched-control study across six quantile-based tasks, and resource-disjoint, display-transfer comparisons against simple defaults. The supplied counterexamples and minimax formulation are applications of elementary identifiability and decision principles, not new mathematical theorems. The intended advance is a reproducible stress test of claims made for computational chart selection.

Results

A declared task connects the image to the score

For samples x and y , the evaluator computes

$$\Delta_k = Q_y(p_k) - Q_x(p_k), \quad p_k \in \{0.10, 0.25, 0.50, 0.75, 0.90\}. \quad (1)$$

These five equally weighted differences describe the lower, middle and upper central portions of two marginal distributions. They do not estimate diagnostic accuracy, a causal treatment effect or within-person change. The empirical quantiles use linear interpolation.

Let c identify a chart family and setting, and e a resolution–observer environment. The renderer produces an image $I_{c,e,r}$ at placement r . The decoder receives only that image and public calibration metadata A_c , including chart type, panel bounds and measurement limits. It returns estimated group quantiles $\tilde{Q}_{g,c,e,r}(p_k)$. Observations, target quantiles and stored summary values are unavailable to the decoder. Define

$$L_{c,e,r} = \frac{1}{5s} \sum_{k=1}^5 |\tilde{Q}_{y,c,e,r}(p_k) - \tilde{Q}_{x,c,e,r}(p_k) - \Delta_k|, \quad (2)$$

$$\bar{L}_{c,e} = \frac{1}{4} \sum_{r=1}^4 L_{c,e,r}, \quad S_{c,e} = \frac{100}{1 + \bar{L}_{c,e}}, \quad (3)$$

$$s^2 = \frac{(n_x - 1)s_x^2 + (n_y - 1)s_y^2}{n_x + n_y - 2}. \quad (4)$$

All candidates for a comparison share the same pooled within-group scale s . A recorded one-source-unit fallback is retained for zero scale. This special case is not unit invariant. A score of 80 means an average absolute contrast error of $0.25s$, not 80% correct readers or retained information.

Since $dS/dL = -100/(1 + L)^2 < 0$, maximizing S is exactly equivalent to minimizing the declared loss. The reciprocal transformation adds interpretability, not information or mathematical novelty. Equal losses remain equal after any strictly monotone transformation. We retain computational ties within absolute loss tolerance 10^{-12} and report winning margins rather than manufacturing uniqueness.

Information eligibility precedes numerical ranking

Let $E_c(D)$ denote the complete image and public calibration supplied by a chart for data D , and let $T(D)$ denote the requested target. A necessary condition for exact recovery is

$$E_c(D_1) = E_c(D_2) \implies T(D_1) = T(D_2).$$

If two inputs have identical encoded information but different targets, any decoder must give the same answer to both. For a scalar target under absolute loss, at least one of its two errors is therefore at least $|T(D_1) - T(D_2)|/2$. This elementary observation separates missing information

from imperfect pixel extraction. Lossy summaries such as mean bars remain diagnostic negative controls: an occasional win on a near-zero target does not make them sufficient distributional representations.

For example (Fig. 1), take paired observations $x = (0, 1, 2)$. Pairing them with $y = (0, 1, 2)$ produces median change zero; pairing them with $y = (1, 2, 0)$ produces median change one. The marginal distributions, common axes and marginal chart inputs are identical. A chart that retains only those marginals cannot identify the median within-person change, regardless of its resolution or decoder. We therefore reject that task for the implemented marginal representations rather than report a misleading score. Means of paired changes are a different case: their average equals the difference of marginal means. The restriction concerns targets that actually depend on pairing.

Strong baselines expose the value of task-matched displays

The expanded benchmark evaluates 23 settings across nine families (Table 1). The original seven families comprise mean bars, dot plots, box plots, variability intervals, histograms, probability heatmaps and violins. Two new baselines address the distributional task directly: an empirical cumulative distribution function (ECDF) and five quantile markers. The latter shows exactly the group summaries needed for the target; it is a deliberately strong task-matched control, not a novel visualization. The renderer may use data to construct the chart, but the decoder must recover its values from pixels. This distinction prevents target leakage into recovery while allowing task-appropriate encodings.

Table 1: **Candidate settings and the equal-budget control.** Fixed settings are chosen before the expanded results are inspected. The original 15-family contextual catalogue is retained in the Supplement; it is not a claim that all 15 support this task.

Family	Full search	One-setting control
Bar chart	Group means	Group means
Dot plot	Mean dots; opaque raw dots	Raw dots
Box plot	Minimum, quartiles, median, maximum	Same
Interval plot	Five-number variability display	Same
Histogram	4, 8, 16, 32, 64 bins	16 bins
Heatmap	4, 8, 16, 32, 64 probability bins	16 bins
Violin plot	8, 16, 32, 64, 128, 256 density grid points	64 points
ECDF	Full empirical cumulative step function	Same
Quantile plot	Five target group quantiles	Same

The primary arrays contain 127 feature comparisons from nine public resources.^{10–18} The reference condition is the original threshold observer at 256 pixels per group panel, averaged over four placements. The new implementation reproduces every original reference-setting loss, including failures, for the 21 existing settings. This parity check isolates the added comparisons from changes to the historical calculation.

The median best reference score is 99.172. The fixed quantile and ECDF baselines have median scores 99.167 and 97.826, respectively; the median best score among the original seven families is 96.414. The quantile baseline beats that seven-family optimum in 124 of 127 comparisons, and the ECDF does so in 76. Table 2 reports adaptive family scores and wins under an equal setting budget. The full score distributions and original equal-budget analysis are retained in Supplementary Fig. S1. Family wins are descriptive development outcomes, not independent replications of superiority.

Table 2: **Reference results across 127 comparisons.** “Wins” permits all 23 settings; “Fixed wins” permits one setting per family. Median S and group error G use each family’s best valid reference setting. Failed settings are excluded from selection, not assigned favorable values.

Family	Wins	Fixed wins	Median S	Median G	Valid
Quantile plot	122	123	99.17	0.014	127
Heatmap	2	1	96.18	0.046	127
ECDF	2	2	97.83	0.023	127
Violin plot	1	0	92.91	0.131	127
Interval plot	0	0	69.94	0.260	127
Histogram	0	0	94.98	0.057	127
Dot plot	0	1	95.58	0.040	127
Box plot	0	0	70.59	0.260	119
Bar chart	0	0	75.10	0.671	122

Six tasks and a stronger control test incremental value

The five-quantile result motivates an explicitly retrospective extension. We retain the nine fixed configurations and add a 19-marker quantile control. Without that additional control, asking for 19 quantiles would disadvantage a display containing only five. Figure 2 reports the complete winning memberships. The two marker budgets are reported as separate configurations; this experiment is not described as an equal tuning budget per family.

Let $d_p = Q_y(p) - Q_x(p)$. The six targets are the median contrast $d_{.5}$; interquartile-spread contrast $d_{.75} - d_{.25}$; unnormalized tail-asymmetry contrast $d_{.9} + d_{.1} - 2d_{.5}$; upper-tail-extent contrast $d_{.95} - d_{.5}$; the original five contrasts; and 19 contrasts at $p = .05, .10, \dots, .95$. Each scalar loss is absolute target error divided by s ; vector losses average coordinatewise absolute errors before division. These are six related distributional questions, not a comprehensive taxonomy of scientific tasks. In particular, they do not test multimodality detection, causal inference or paired change. No weighted score combines their losses.

All charts are rendered unchanged across tasks. Five-marker values are interpolated linearly between marks and constantly beyond their range; the 19-marker control directly encodes all probabilities used here. ECDF and other distributional decoders recover the requested quantiles from their respective image representations. The dense control has no omitted-target error for these tasks by construction, but still incurs pixel recovery error. This is an intentionally demanding control, not evidence of a novel chart design.

We compare six chart policies: fixed five-marker, fixed 19-marker and ECDF displays; a resource-disjoint task default; case-specific reference selection; and a test-condition oracle used only as a descriptive bound. For each task and observer, the default selects one configuration using the other eight resources at 256 pixels, with equal weight per resource. The case-specific policy instead selects the best configuration for that case at 256 pixels. Both choices are frozen and evaluated at 384 pixels and four new placements. No test-display losses enter either selection. A seventh, non-graphical control predicts a zero target for each task; it tests whether a small recovery error adds value beyond assuming no difference. Because the resources and display conditions were already involved in development, this is a retrospective transfer experiment, not an untouched validation study.

Table 3 reports resource-balanced transfer loss under the threshold observer. Positive gain means lower loss for case-specific selection than for the resource-disjoint task default. The intervals resample nine paired resource summaries, not 127 independent features. They are exploratory descriptive intervals without simultaneous multiplicity coverage; they do not establish practical human benefit. Both observers, fixed controls, failures and computational tolerance counts are supplied in the Supplement.

Table 3: **Does case-specific selection improve on a task default at a new display condition?** Each resource has equal weight. Q19 is the fixed 19-marker control. Gain is default loss minus case-selection loss; negative values favor the simpler default. Intervals use 5,000 paired resource bootstrap resamples. All losses are in pooled-SD units.

Task	Q19	Task default	Case selection	Gain (95% interval)
Median	0.0067	0.0063	0.0063	-0.0000 [-0.0009, +0.0011]
Interquartile spread	0.0088	0.0086	0.0091	-0.0006 [-0.0013, +0.0001]
Tail asymmetry	0.0159	0.0161	0.0142	+0.0019 [+0.0006, +0.0032]
Upper-tail extent	0.0096	0.0096	0.0098	-0.0002 [-0.0005, +0.0001]
Five quantiles	0.0063	0.0063	0.0064	-0.0000 [-0.0001, +0.0000]
19 quantiles	0.0062	0.0062	0.0063	-0.0001 [-0.0003, +0.0000]

Case-specific selection does not consistently improve on the task default. For tail asymmetry, its threshold-observer gain is 0.00189 pooled SD (descriptive 95% interval 0.00060 to 0.00315); the corresponding alpha-observer gain is 0.00034 (-0.00106 to 0.00198). For the other five threshold-observer tasks, the point estimate favors the default. Under the alpha observer, selection increases mean loss for spread and upper-tail extent, with descriptive intervals below zero. No evaluated policy has a failed transfer case, although 168 candidate-case-task-environment aggregates are invalidated by failed placements. These results do not support a general advantage for choosing a separate chart for every case.

At the transfer display, the 19-marker control belongs to the dense-task winning set in 127 cases under the threshold observer and 126 under the alpha observer. The earlier dense-task ECDF advantage therefore does not survive adding the directly matched dense control. This is evidence about comparator adequacy, not a human preference for more quantile markers.

Separate omitted information from raster error

For the direct controls, let T_c^* be the target implied by exact plotted markers and the declared interpolation rule before rasterization, and let $\tilde{T}_c$ be recovered from pixels. Define $L_{\text{enc}} = \|T_c^* - T\|_1/(ms)$ and $L_{\text{pix}} = \|\tilde{T}_c - T_c^*\|_1/(ms)$ for a target with m coordinates. The triangle and reverse-triangle inequalities give

$$|L_{\text{enc}} - L_{\text{pix}}| \leq L \leq L_{\text{enc}} + L_{\text{pix}}.$$

These are bounds, not an additive decomposition: encoding and pixel errors may cancel. The 19-marker control has $L_{\text{enc}} = 0$ for every tested task. The five-marker control has zero encoding error for the median, spread, asymmetry and five-quantile tasks, but can omit information needed for the upper-tail and dense tasks. Figure 4 uses the same prespecified WDBC feature across displays; it does not select a different favorable case for each chart.

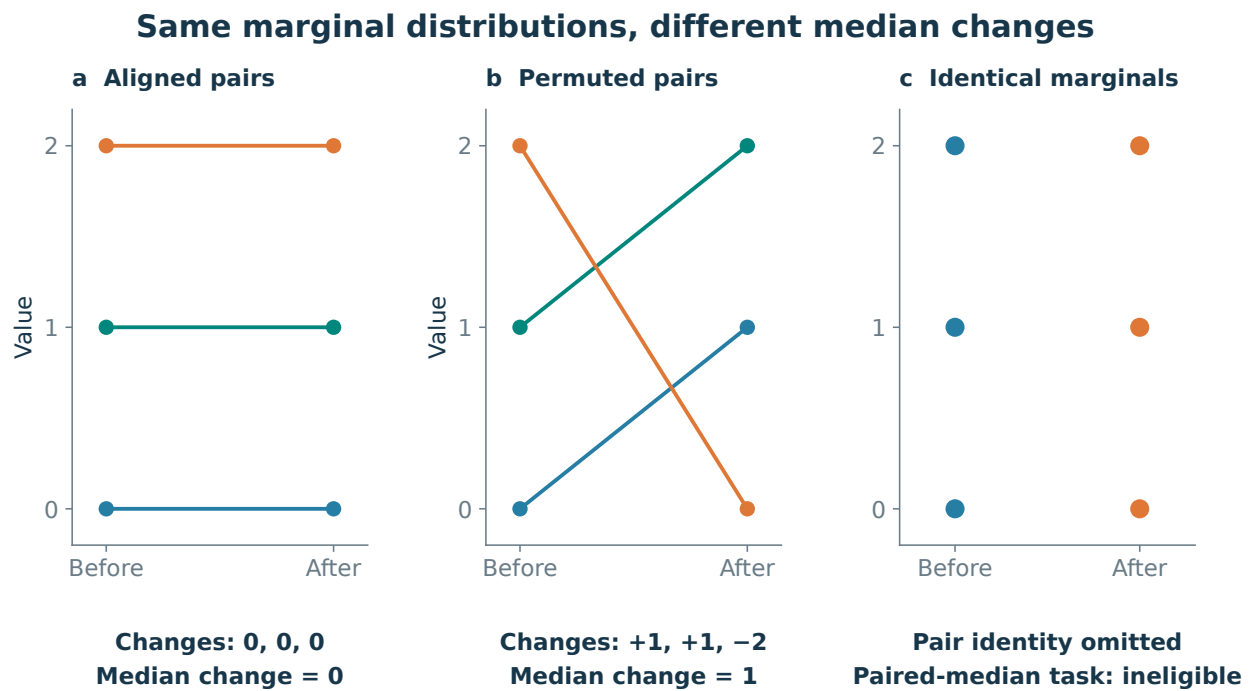
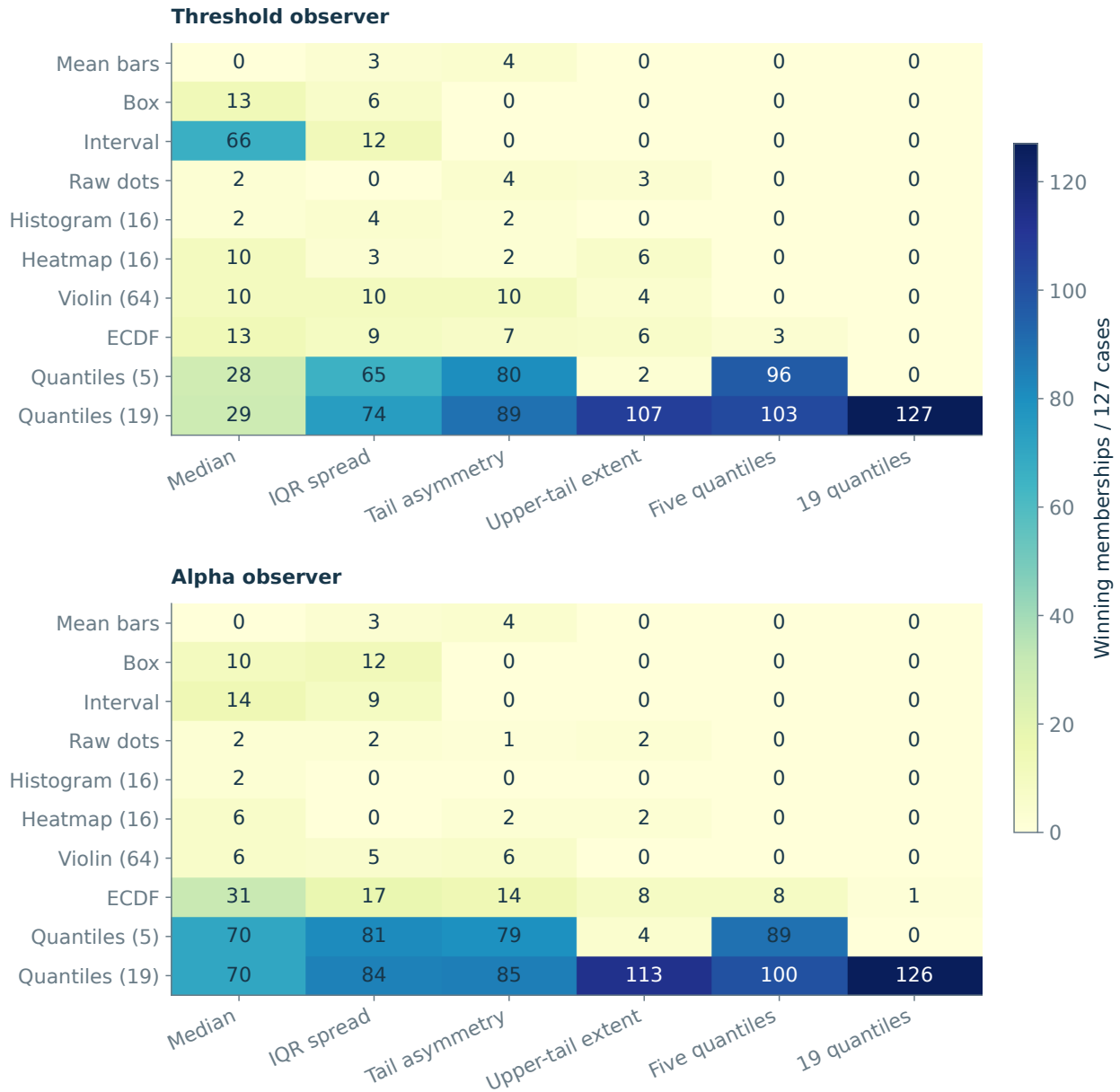


Figure 1: **Information lost before rendering cannot be recovered by ranking charts.** Two explicitly synthetic pairings have the same group marginals but median changes of zero and one. Connections preserve the information needed for the paired task; separate marginal summaries omit it. The witness concerns median within-person change, not the difference of marginal medians or the mean paired change. No empirical patient result is represented.

Who wins after adding the task-matched control?



384-pixel evaluation. Columns retain ties. Two quantile budgets are separate configurations.

Figure 2: **Task dependence with a matched dense control.** Winning configuration memberships at 384 pixels, averaged over four placements, are reported separately for each task and observer. Scientific ties are retained, so a column can exceed 127 memberships. Five- and 19-marker controls are separate configurations. A change in winner does not by itself establish an advantage for adaptive selection, which is tested in Fig. 3.

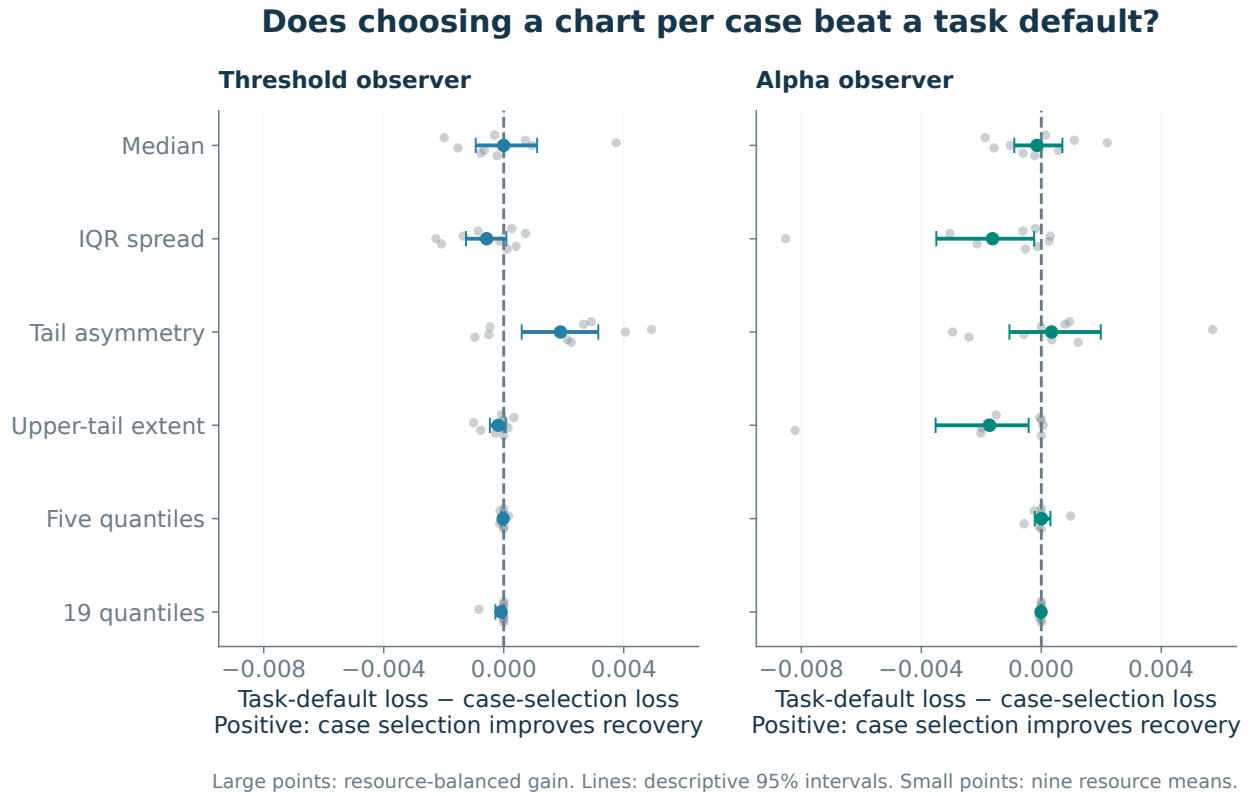
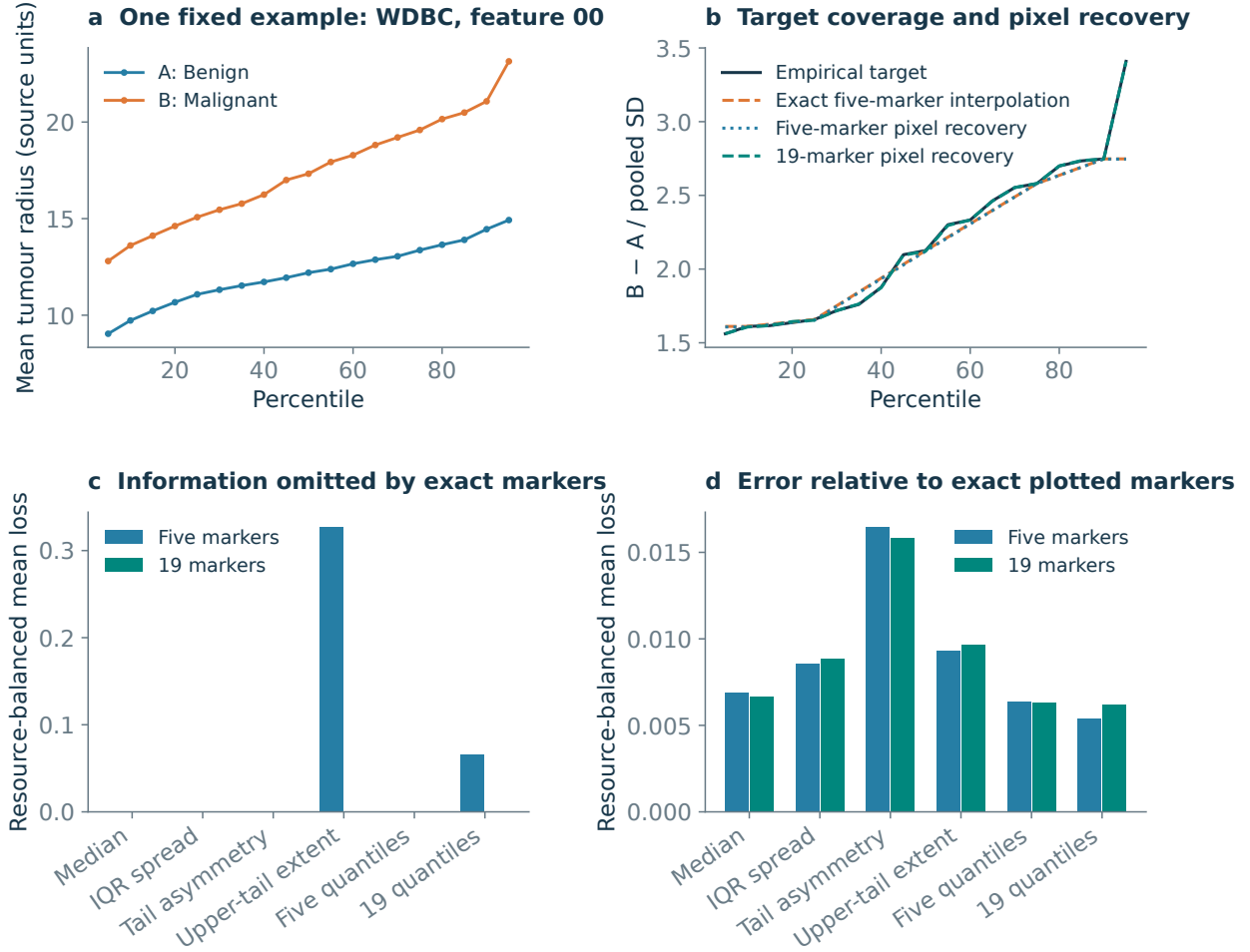


Figure 3: **Incremental benefit must survive transfer.** Points show task-default loss minus case-specific-selection loss at 384 pixels; positive values favor case-specific selection. Lines are descriptive 95% paired bootstrap intervals over the nine resource means. Selections were made at 256 pixels. Neither these intervals nor the two computational observers establish human performance. Tasks are analyzed separately, with no combined utility score.

Separate what the chart omits from what pixels distort



Panels c-d: all 127 cases, nine equally weighted resources. Error magnitudes need not add because signs can cancel.

Figure 4: A task change can expose omitted information without creating a new selector. The first WDBC feature by canonical key is shown in all panels. Exact five-marker interpolation, exact 19-marker values, and the corresponding recovered curves distinguish target coverage from raster error. Lower panels summarize encoding and pixel error across all nine resources at 384 pixels under the threshold observer. Their sum is an upper bound on total error, not an equality.

Group fidelity and trivial baselines qualify a high score

Contrast recovery can hide errors in both groups. We therefore report, without combining it into a new weighted score,

$$G_{c,e} = \frac{1}{4} \sum_r \frac{1}{10s} \sum_{g \in \{x,y\}} \sum_k |\tilde{Q}_{g,c,e,r}(p_k) - Q_g(p_k)|. \quad (5)$$

By the triangle inequality, $\bar{L}_{c,e} \leq 2G_{c,e}$, but a small L does not imply a small G . Adding the same error to both recovered groups yields zero contrast error while distorting both distributions. This is a mathematical limitation of the objective, not a decoder-specific anomaly.

A second control predicts zero difference at every quantile, with $L_0 = (5s)^{-1} \sum_k |\Delta_k|$. We report $1 - \bar{L}/L_0$ when $L_0 > 0$; negative values mean that decoding performs worse than this trivial prediction. When $L_0 = 0$, the relative skill is undefined and remains missing. Full group errors, zero-baseline losses and relative skills accompany every family result. Near-null contrasts can have high absolute scores even when the image adds little beyond predicting no difference.

A more basic limit occurs before rasterization: different distributions can have the same displayed summary. Under a common fixed axis, two mean-only displays may be identical despite different quantile targets. No decoder receiving the same image and metadata can distinguish those cases perfectly. The Supplement gives an explicit equal-mean, equal-variance construction and a lower bound. This motivates strong baselines that actually encode the declared target.

Robust choices complement precise-looking reference ranks

The best family changes in 13 of 127 comparisons between 128 and 512 pixels. The alternative alpha-based observer changes 14 reference winners at 256 pixels. These observers share chart geometry and design assumptions; their comparison is algorithmic sensitivity, not independent perceptual validation. Tight measurement axes are evaluated separately, excluding mean bars from both sides because removing zero can invalidate their baseline.

The median best-versus-second-best family score margin is 1.272; 2 comparisons have margins below 0.1 points. Reporting three decimals therefore does not imply that the differences are perceptible or stable. For an explicit excess-error budget ϵ , define

$$\mathcal{C}_\epsilon(e) = \{c : \bar{L}_{c,e} \leq \min_j \bar{L}_{j,e} + \epsilon\}. \quad (6)$$

We report family membership at $\epsilon \in \{0.001, 0.005, 0.01, 0.025\}$ pooled standard deviations. At $\epsilon = 0.01$, 51 comparisons admit multiple families. These are declared computational tolerances, not empirically validated thresholds of human equivalence.

If a single unchanged setting must operate across environments $\mathcal{E}$, define

$$R_c = \max_{e \in \mathcal{E}} \left[\bar{L}_{c,e} - \min_{j \in \mathcal{C}_{\text{valid}}(e)} \bar{L}_{j,e} \right], \quad \mathcal{C}_{\text{robust}} = \arg \min_{c \in \cap_e \mathcal{C}_{\text{valid}}(e)} R_c. \quad (7)$$

The six selection environments combine two observers with three resolutions. A candidate must decode at every placement in every environment to qualify as robust. The environment-specific comparator may use any setting valid in that environment. This is a standard minimax-regret construction applied to a declared chart benchmark; it is not a claim of a new optimization theorem. Its training-envelope advantage follows from its definition and must not be mistaken for out-of-environment validation.

We therefore freeze each comparison’s reference and minimax settings and transfer them to 384-pixel panels with placements 0.125, 0.375, 0.625, 0.875, which are excluded from selection. The fixed quantile plot is transferred alongside them. Supplementary Fig. S2 and the accompanying tables report resource-balanced errors for both observers, including any failure. At the new display condition,

resource-balanced mean losses for the fixed quantile, reference-selected and minimax-selected charts are 0.00638, 0.00646 and 0.00655 for the threshold observer, and 0.00339, 0.00383 and 0.00351 for the alpha observer. Thus minimax selection does not consistently improve on reference selection, and the simple fixed quantile baseline has the lowest mean loss for both observers. This separates retrospective optimization from performance at a new display condition. It does not test an unseen medical population.

Resource transfer and simulation extend the development audit

A leave-one-resource-out analysis selects one global configuration using mean loss on the other eight resources, weighting resources equally. The selected setting is then evaluated on every comparison in the omitted resource without case-specific tuning. Its resource-balanced held-out loss is 0.0093. The chosen configurations and comparison-level outputs are reported in the Supplement. Because all resources were available during development, this is retrospective resource-disjoint evaluation, not untouched external validation.

Seventy-two synthetic comparisons span null, location-shift, scale-change, skewed, mixture and heavy-tailed regimes; sample sizes are 20, 100 and 500 per group, with four replicates per cell. The nine fixed settings, seeds and generation rules are declared before the synthetic run. Scores are evaluated against each generated sample’s empirical target, not its population quantile difference. These modest simulations expose controlled failure modes; they do not establish universal performance across distributions or replace human observations.

Data-specific sensitivities omit the two flagged CKD potassium values without replacing them and repeat results after excluding the 16 retinopathy features with unresolved participant linkage. The Parkinsons comparisons remain outside the primary analysis because parsed identifier groups conflict with documented participant counts. No data correction is chosen to improve chart scores.

Discussion

The central result is a test of incremental value, not a claim that every dataset needs a different chart. A five-quantile target predictably favors a display of those five quantities. Expanding the task requires expanding its matched control as well; otherwise an apparent winner reversal may simply reveal information omitted from the earlier baseline. The six-task study and display-transfer comparison make this distinction measurable. They support conditional computational conclusions, while placing limits on claims of a general-purpose recommender.

The elementary eligibility criterion also separates questions that chart optimization cannot resolve. A marginal representation cannot recover a pairing-dependent target merely because its marginal quantiles are accurate. The same logic applies whenever distinct inputs collapse to the same encoded summary. An operational selector should reject an unsupported target and request an appropriate representation. It should then compare any admissible optimization with a strong default and report near-equivalent choices, failures and sensitivity rather than a uniquely authoritative rank.

These results do not establish that a 19-marker display is easier for a person to use than an interval or ECDF. It directly includes all summaries used in this benchmark, and a calibrated machine can exploit that alignment. Estimation effort, visual clutter, task switching and reader expertise are absent from the objective. Adding arbitrary penalties for those factors would not supply the missing evidence. A claim that optimization improves scientific communication requires those outcomes to be measured.

Decoder dependence remains a central limitation. The threshold observer and alpha observer use known calibration and family-specific recovery rules. Neither is independently trained or validated against human behavior. A probability heatmap’s calibrated intensities are machine-readable without

establishing perceptual accessibility.¹⁹ Opaque-dot overlap can impair an area-based decoder without implying that a different algorithm or person fails identically. Agreement between the two observers is therefore reassuring only within this narrow class of algorithms. Disagreement reveals fragility; it does not identify which observer better models a reader.

The benchmark also separates three kinds of uncertainty. Placement and resolution checks address rendering sensitivity. Resampling resource-level summaries describes variation among the nine convenience resources. Sampling uncertainty in population quantile differences is a different problem and is not estimated by either procedure. Participant-level source aggregation reduces repeated-measure inflation in HAR, BIDMC and wrist data, but features within a resource remain correlated. Small wrist cohorts, unresolved retinopathy linkage and questionable CKD values constrain biological interpretation. We study empirical image recovery, not disease effects inferred from these resources.

The zero-inclusive axis and SD normalization are deliberate but imperfect choices. Including zero can waste space for non-bar encodings; tight-axis results expose this dependence. Pooled SD is sensitive to extremes. Changing to another positive scale shared across candidates would rescale their losses without changing the ranking for that comparison; it could change pooled summaries across comparisons. Zero-scale fallback remains source-unit dependent. We disclose these limits rather than presenting a score threshold as a universal quality certificate.

Human validation remains the decisive next step for claims about readers. The included prospective protocol now treats selection versus a frozen task-matched default as its primary policy comparison and includes the dense control; it is a study plan, not completed evidence. A prospective experiment must freeze stimuli and computational predictions, use untouched cases, randomize display assignment, and measure estimation error alongside time and confidence. It should test whether the score adds predictive value beyond chart family and direct quantile baselines, accounting for repeated participants and cases. The accompanying protocol specifies these analyses and provides planning tools, but supplies no recruited participants, responses or approval claims. The present evidence supports computationally auditable choices for a fixed task; it does not establish improved human interpretation or clinical decisions.

Methods

Source arrays and units of analysis

The original prepared arrays are retained: WDBC 30 comparisons, Cleveland five, heart failure six, CKD 11, ILPD nine, retinopathy 16, HAR 40, BIDMC five and wrist five. HAR uses subject/activity medians from 30 participants. BIDMC summarizes readings within 46 source patients. Wrist exercise uses early/late ECG-reference summaries within eight participants.^{20;21} Missing observations and transformations remain recorded in the preparation pipeline. Paired resources are explicitly labeled, but the target compares marginal quantiles. Parkinsons contributes 22 exploratory comparisons only in the archived analysis.

Rendering and recovery

The common reference interval includes zero and extends four times the larger group Scott bandwidth beyond the observed extremes, with a bandwidth floor of 0.01s. Each group occupies a separate square panel of side 128, 256 or 512 pixels, with common measurement limits. Images are drawn at fourfold resolution, reduced with Lanczos filtering and blurred by a 0.5-pixel Gaussian. Original mark sizes and colors are unchanged. Four vertical placements are evaluated; raw-dot jitter also changes deterministically, whereas heatmap placements repeat the same strip. These are specified stress conditions, not independent random reader samples.

The original observer extracts thresholded colored or black pixels. The alternative projects RGB deviation from white onto the known group-color direction, clips the resulting alpha estimate to $[0, 1]$, uses an alpha threshold of 0.2 for geometric masks and integrates alpha above 0.05 for area recovery. Heatmap recovery averages a local 3×3 neighborhood in the alternative observer. Both observers retain known layout and family-specific geometric assumptions. The Supplement and source code give exact rules and failure conditions.

The ECDF is a two-pixel step curve with cumulative probability on the horizontal axis and measurement on the vertical axis. Decoding estimates the ordinate at each target probability column. Finite-sample ECDF inversion differs from linear-interpolated empirical quantiles, a retained source of error. Quantile plots place radius-two-pixel markers at the five target probabilities; decoding uses local pixel-mass centroids, with no numerical target values in metadata. These baselines occupy the same panel area as the other charts. The six-task extension adds 19 markers at probabilities .05, .10, . . . , .95, with the same radius and extraction window. Recovered quantiles are monotonized by cumulative maximum; consequently, a denser sequence can also change the recovered values at shared probabilities when adjacent estimates violate monotonicity. The extension evaluates 256- and 384-pixel panels with the placements specified above.

Selection and validation

Loss is averaged across four placements before transformation. A failed placement invalidates the setting in that environment. Family summaries use the minimum valid loss within family. Exact ties use absolute tolerance 10^{-12} ; deterministic display order does not imply scientific superiority. Equal-budget controls use the fixed settings in Table 1. The minimax rule requires a configuration to succeed in all six selection environments. The new-resolution transfer uses fixed selections and new placements without retuning.

Resource-disjoint selection requires a candidate to decode every training comparison, averages feature loss within each training resource and then weights resources equally. The selected configuration’s held-out failures are retained. Descriptive bootstrap intervals for each family’s loss difference from the quantile baseline resample resource means 5,000 times with a fixed seed; invalid comparisons are omitted and counts disclosed. These intervals concern the sampled resource summaries and are not population-generalization guarantees.

Both protocols are retrospective extensions, not preregistrations. In the earlier audit, a deterministic fixture failure motivated increasing the new ECDF stroke from one to two pixels before the final run; the stopped initial run supplies no reported results. Protocol, source and input hashes are recorded. The earlier checks cover all 23 settings and both observers, including blank-image rejection, pixel dependence, positive unit rescaling and metadata restrictions. Original reference losses are independently compared with the supplied archived CSVs. The six-task verification additionally recomputes every target from saved recovered quantiles, verifies resource exclusion for all 108 default-selection folds, and checks all 9,144 chart-policy transfer values. A zero-target control is evaluated separately and documented as a post-run diagnostic. Source code, per-trial recovered values, aggregate outputs and figure-generation scripts are included.

Data availability

The source resources are available through the cited UCI and PhysioNet records. The accompanying reproducibility archive contains prepared arrays, source copies where supplied, acquisition records, attribution, transformations and checksums. Source-specific terms are documented in `DATA_PROVENANCE.md`; original licenses are retained. Synthetic inputs are explicitly identified and included. No human-reader data were collected.

Code availability

The complete version 3 analysis, with the version 2 audit preserved, accompanies this manuscript in the reproducibility archive. Running `python reproduce_v3.py` executes the six-task benchmark, policy comparisons, verification and current figure generation. The earlier audit is reproduced separately with `python reproduce_strengthened.py`. Pinned dependencies, deterministic seeds, source hashes and per-trial CSVs are supplied. The associated project repository is <https://github.com/Elgendi/two-sample-chart-selection>; the accompanying archive identifies the exact code underlying this version.

Author contributions

M.E. conceived the study, developed the methodology, performed the analyses, interpreted the results, and wrote and revised the manuscript.

Competing interests

M.E. serves on the editorial boards of Communications Medicine and npj Biosensing. The author declares no other competing interests.

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